Business Data Mining Homework Session 4

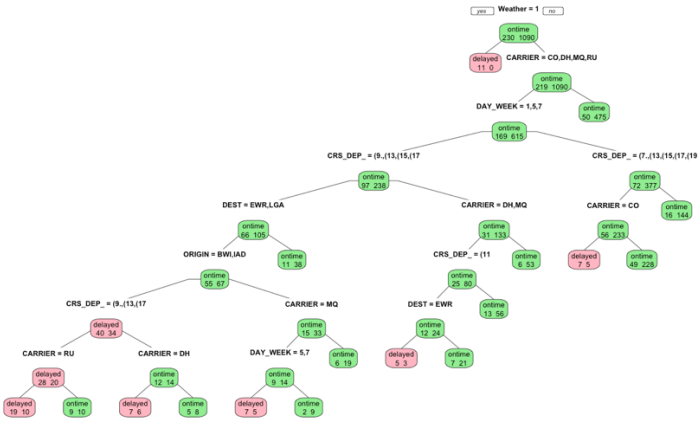
9.2 Predicting Delayed Flights. The file FlightDelays.csv contains information on all commercial flights departing the Washington, DC area and arriving at New York during January 2004. For each flight, there is information on the departure and arrival airports, the distance of the route, the scheduled time and date of the flight, and so on. The variable that we are trying to predict is whether or not a flight is delayed. A delay is defined as an arrival that is at least 15 minutes later than scheduled.

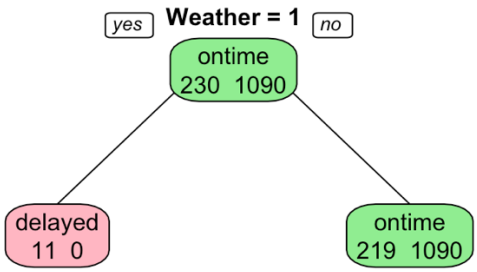
Data Preprocessing. Transform variable day of week (DAY_WEEK) info a categorical variable. Bin the scheduled departure time into eight bins (in R use function cut(). Use these and all other columns as predictors (excluding DAY_OF_MONTH). Partition the data into training and validation sets.

training set size: 1320
validation set size: 881
Proportion of delayed flights in training set: 0.1977273
The proportion of training set is 60%, and the proportion of validation data is 40%.

a. Fit a classification tree to the flight delay variable using all the relevant predictors. Do not include DEP_TIME (actual departure time) in the model because it is unknown at the time of prediction (unless we are generating our predictions of delays after the plane takes off, which is unlikely). Use a pruned tree with maximum of 8 levels, setting cp = 0.001. Express the resulting tree as a set of rules.

We use 7 independent variables, which are "CRS_DEP_TIME", "CARRIER", "DEST", "DISTANCE", "ORIGIN", "Weather", "DAY_WEEK". The left one is unpruned tree and the right one is pruned tree.





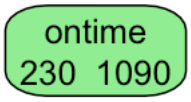
Rules of pruned tree are as follows.		
2	0.00 flight status when Weather is 1	Cover 0.42%
3	0.83 flight status when Weather is 0	Cover 99.58%

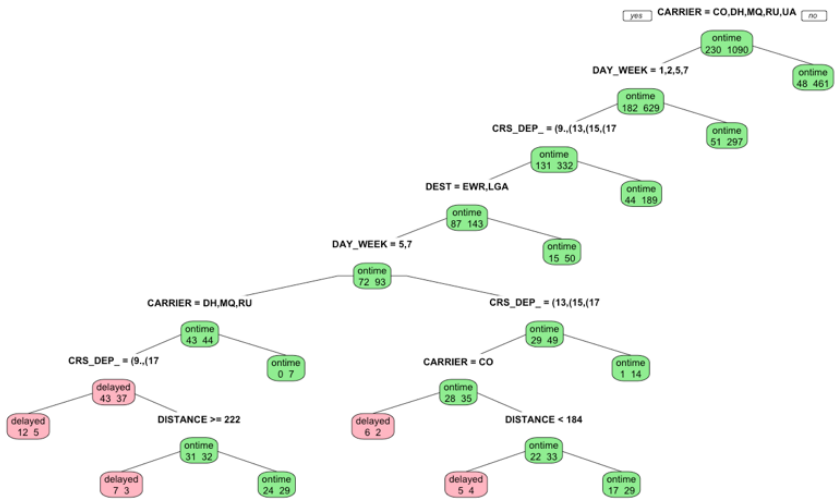
When the Weather is 1, the pruned tree defaults the flight status is ‘delay’. there are 11 observations, and the total observations are 2640 (11+0+219+1090+230+1090). So, the cover rate is 0.42%.
When the Weather is 0, the pruned tree combined two nodes. The total observations are 2629 (230+1090+219+1090), the observations of ‘ontime’ are 2180 (1090+1090). So, the rate of not delay is 0.83 (2180/2629). And the cover rate is 99.58% (2629/2640).

b. If you needed to fly between DCA and EWR on a Monday at 7:00 AM, would you be able to use this tree? What other information would you need? Is it available in practice? What information is redundant?

Yes, we would be able to use this tree for predicting a Monday 7:00 AM flight from DCA to EWR. All required predictor variables can be provided at booking time. Flight route (DCA-EWR), schedule time (7:00 AM), day of week (Monday) are all known. Carrier selection and distance are fixed and known. However, the variable of Weather requires access to weather prediction data. The accuracy depends on reliable weather information availability. Therefore, I think this model is available in practice. It is easy to understand and use simple and accessible information.
DISTANCE is highly correlated with ORIGIN and DEST, so I think these information is redundant.

c. Fit the same tree as in (a), this time **excluding the Weather predictor**. Display both the pruned and unpruned tree. You will find that the pruned tree contains a single terminal node.





The left one is pruned tree, and the right one is unpruned tree.

i. How is the pruned tree used for classification? (What is the rule for classifying?)

When the Weather variable is excluded, the pruned tree predicts all flight status is ‘ontime’ regardless of any other variables. This is the simplest and easiest way to classify. The total observations are 1320 (230+1090). The observations of ‘ontime’ are 1090. Therefore, this rule achieves 82.6% accuracy (1090/1320). However, it completely fails to detect any delayed flights (0% sensitivity). The reason is because the predictive power of the remaining variables is insufficient.

ii. To what is this rule equivalent?

This single-node rule is equivalent to a baseline majority classifier that ignores all predictor variables. Predicting the most frequent class from the training set for all observations. A model with no predictive power beyond the prior probability distribution. Always predicting "ontime" regardless of flight characteristics, carrier, time, or other factors.

iii. Examine the unpruned tree. What are the top three predictors according to this tree?

Carrier, DAY_WEEK, CRS_DEP. Because these 3 variables appear at the highest split levels of unpruned tree. Carrier is the nodes splitter, which shows it is the most important predictor. DAY_WEEK appears at the second split level and CRS_DEP appears at the third split level.

iv. Why, technically, does the pruned tree result in a single node?

When the predictive power of other variables is weak, the error reduction achieved by adding leaf nodes is insufficient to offset the complexity penalty, thus the optimal CP value corresponds to the simplest single-node tree. More complex trees perform worse on the validation set due to overfitting, while the single-node tree actually achieves the minimum generalization error.

v. What is the disadvantage of using the top levels of the unpruned tree as opposed to the pruned tree?

The unpruned tree suffers from poor generalization as it overfits the training data, resulting in reduced predictive accuracy on new data. It produces overly complex rules with high variance, making the model unstable and difficult to interpret. Additionally, it requires higher costs and longer prediction times compared to the pruned tree.

vi. Compare this general result to that from logistic regression in the example in Chapter 10. What are possible reasons for the classification tree's failure to find a good predictive model?

Accuracy of Logistic Regression and Decision Tree Models			
model	accuracy	sensitivity	specificity
Logistic Regression	0.7820658	0.2525253	0.9355783
Decision Tree	0.7990919	0.1060606	1.0000000

The classification tree performs poorly because it over-prioritizes the majority class ("ontime"), essentially ignoring delayed flights with only 10.6% sensitivity. Unlike logistic regression that uses all variables together to capture complex relationships, the tree makes irreversible poor splits early on and struggles with continuous patterns and multi-factor interactions that cause flight delays.